Functional connectivity

An introduction to functional connectivity with fMRI data

Dr. Alessio Borriero

Computational application to cognitive neuroscience

UNIVERSITÀ DI TORINO

Brain connectivity

The idea of a connected brain

has a very long story, rooted in neuropsychological studies

Disconnection syndromes

a brain lesion can impair a function by disrupting interactions, not only by destroying a region that "contains" that function.

Wernicke - 1874
Lichteim - 1885

Aphasia

Gazzaniga -1967

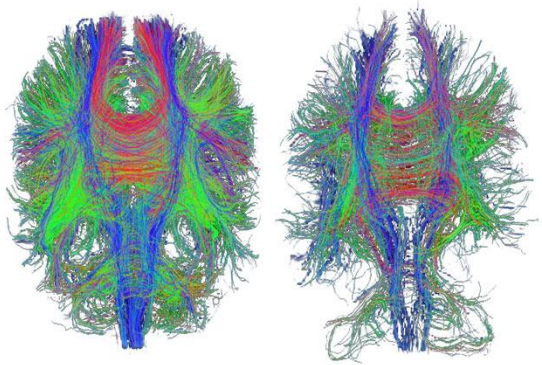
Split brain

Geschwind -1965

"Disconnexion Syndromes in Animals and Man"

Many types of connectivity

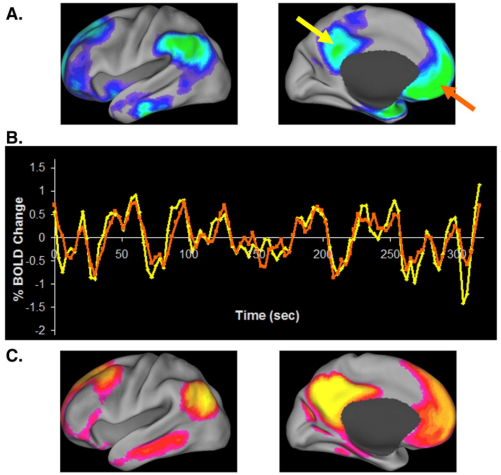
Structural



https://www.umimri.com/dti/

connections between neurons are too “slow” to explain cognitive functions…

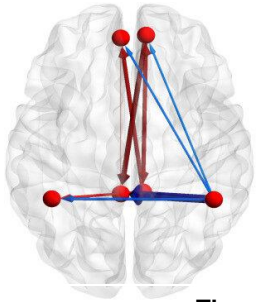
Functional



Raichle, 2007

To study the functional connectivity allows to looks at the emergent properties of the brain

Effective



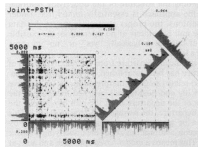
Ting, 2014

Models of causal relations between brain regions or group of neurons

Granger causality

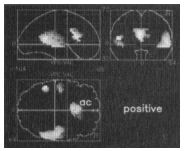
Whole brain computational modeling (e.g. Hopf model)

A brief timeline of studies about functional connectivity



Correlation of spike trains

1989
Aersten et al



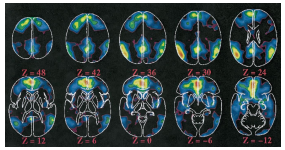
PCA-based FC with PET

1993
Frinston et al.



Seed-based FC with rs-fMRI between motor cortices

1995
Biswal et al.

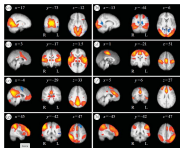


Default-mode network in PET data

2001
Raichle et al

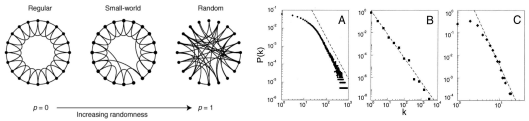
Default-mode network in fMRI data

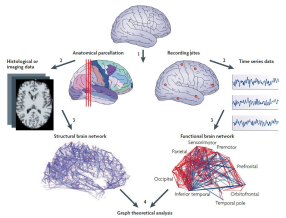
2003
Greicius et al



Resting-state networks

2005
Beckmann et al





Network Theory

1998
Watts & Strogatz

Collective dynamics of small-world networks

1999
Albert & Barabasi

Emergence of scaling in random networks

2004
Sporns et al.

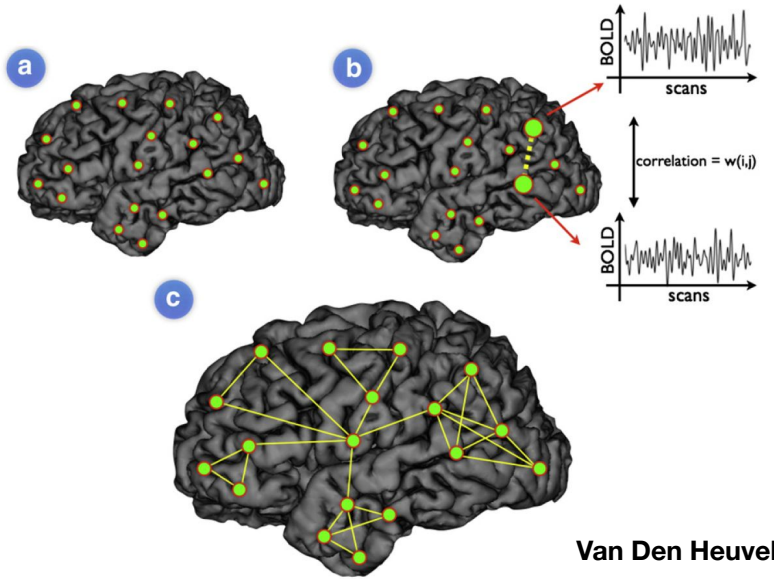
Organization of complex brain networks

2009
Bullmore & Sporns

Graph-based analysis of complex brain networks

What does Functional Connectivity (FC) mean?

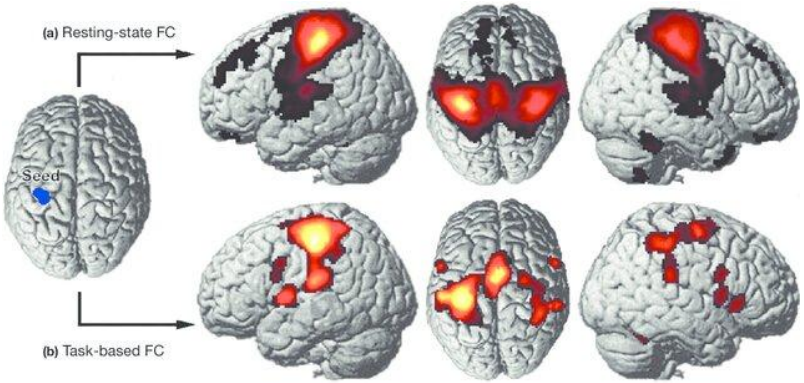
correlation of some kind of neurophysiological signal measured in different brain regions



Van Den Heuvel et al., 2010

Different approaches to FC with fMRI

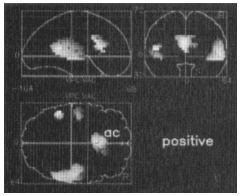
model-dependent
(seed methods)



model-free
(without a priori information)

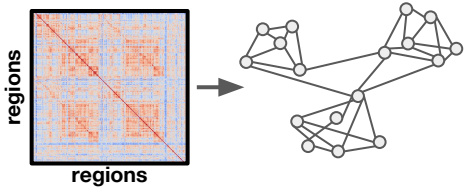
PCA

ICA



clustering methods

network-based analysis



The connectivity is often studied at rest

The incredible amount of information in the activity at rest of the brain at rest was discovered studying the "physiological noise" of the brain (cardiac and respiratory -related signal).

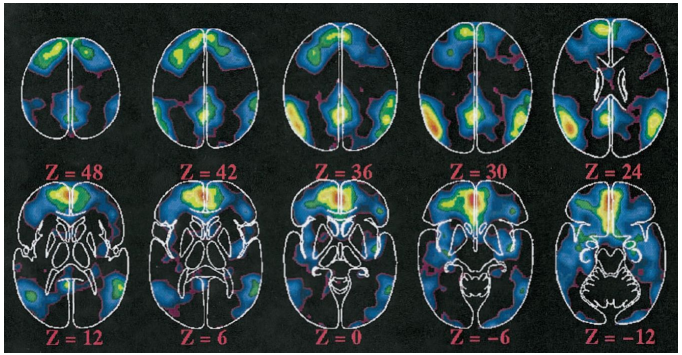
Even removing this kind of noise, some residual spatio-temporal correlation of the signal remains…

Lowe, 2010

The Default-Mode Network

Since the first experimental observation of the brain in the early 20th century researchers has observed that the amount of activity of the brain at rest is nearly the same as during an active task.

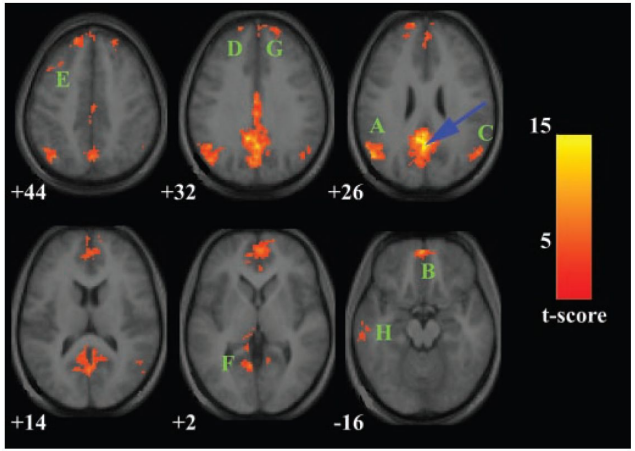
In 1997, Shulman et al. observed the existence of *task-negative network* that seemed to be deactivate during active tasks, like language, motor or visual goal-directed tasks.



The Default-Mode Network

In 2001, Raichle et al. observed the same pattern of activated brain areas in subjects at rest using PET imaging. They coined the term *Default Mode Network* for this particular network.

In 2003, Greicius et al. discover the same network with fMRI resting state data.



The Default-Mode Network

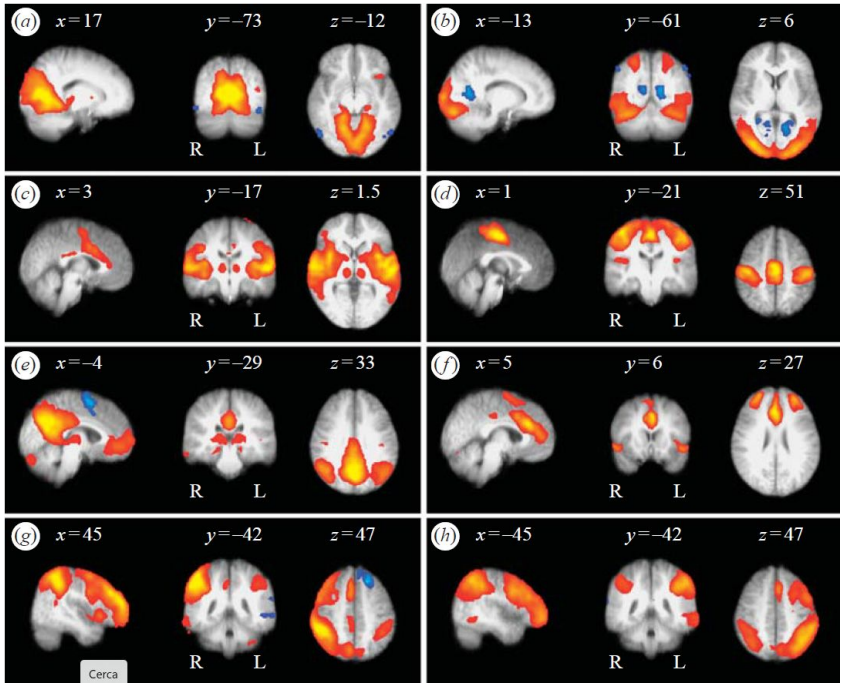
Following this initial works, many studies have addressed the role of the Default Mode Network with respect to representation of the self, thinking about others and remembering the past.

Alteration of the DMN have also been related to clinical disorders such as Alzheimer's disease, autism depression and many others.

The brain region composing the DMN are the medial prefrontal cortex, the posterior cingulate cortex, the precuneus and the angular gyrus.

The Resting State Networks

In 2005, Beckmann et al. Defined a set of resting-state networks performing an ICA analysis over resting-state fMRI data.



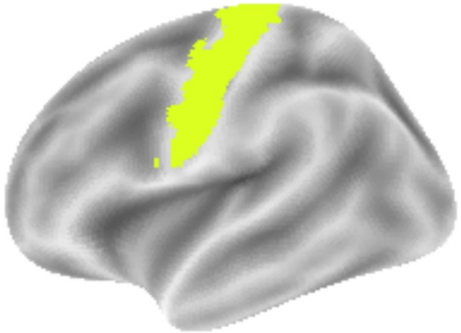
Our lectures about functional connectivity with fMRI

01 - Seed-based FC

02 - Graph analysis applied to resting-state fMRI

01 - Seed-based FC

Just choose a seed region of the brain



And then correlate the average time series of the seed with all the voxels of the brain

It’s simple :)

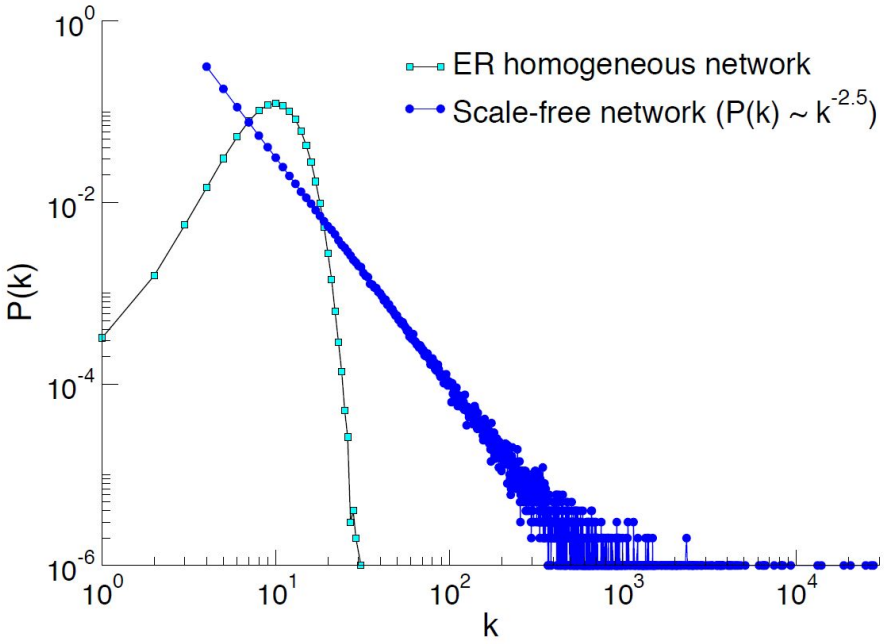
02 - Graph analysis applied to resting-state fMRI

Definitions

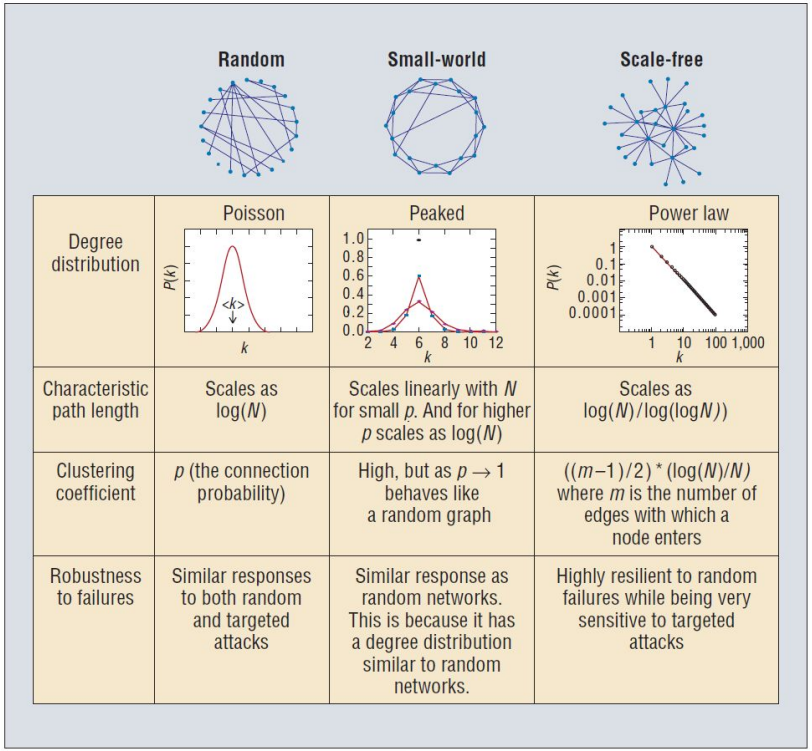
$$G = (V, E) : Graph$$
$$V = \{v_1, \ldots, v_n\} : Set\ of\ vertices$$
$$E = \{\{v_1, v_2\}, \ldots, \{v_n, v_m\}\} : Set\ of\ edges$$
$$A = (a_{ij}) : Adjacency\ matrix,$$
$$where\ a_{ij} = \begin{cases} 0 & if\ \{v_i, v_j\} \in E, \\ 1 & if\ \{v_i, v_j\} \notin E \end{cases}$$

02 - Graph analysis applied to resting-state fMRI

Random graphs, scale-free property and small wordness



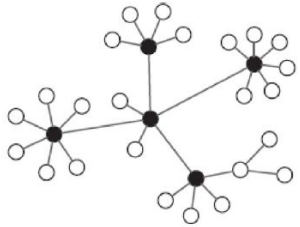
Boronchelli et al., 2013



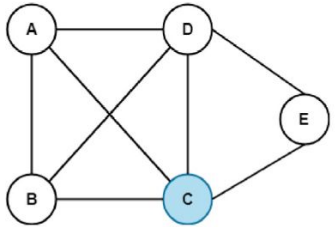
Thadakamalla et al., 2004

02 - Graph analysis applied to resting-state fMRI

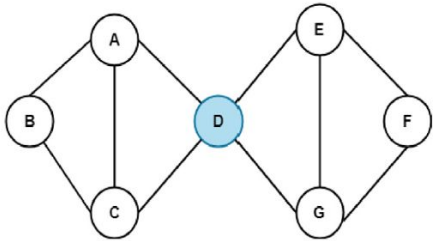
Centrality measures



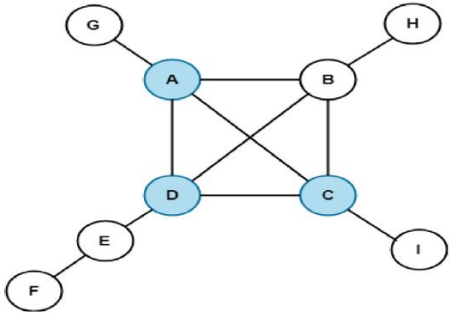
(A) Degree Centrality



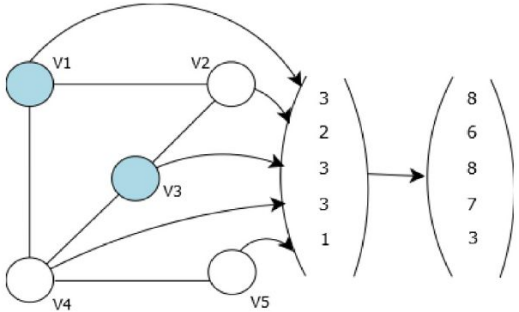
(B) Closeness Centrality



(C) Betweenness Centrality



(D). Jordan Centrality

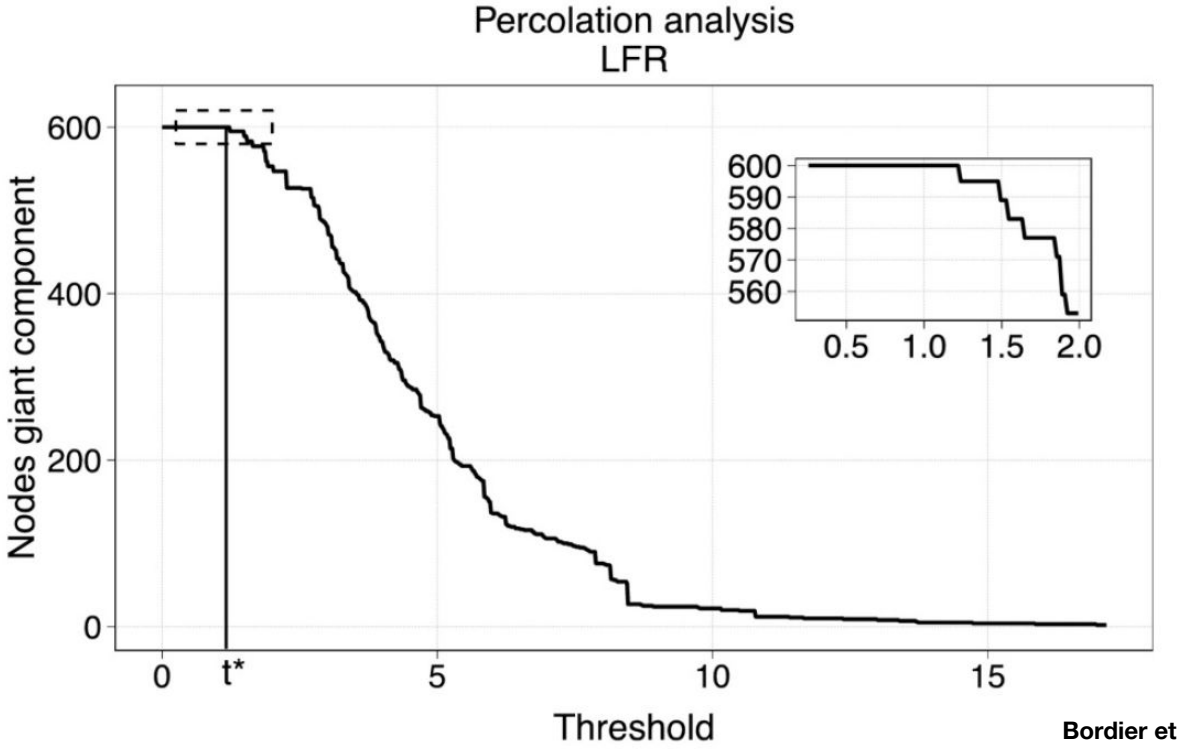


(E) Eigenvector Centrality

Shelke et al., 2019

02 - Graph analysis applied to resting-state fMRI

Percolation



Bordier et al., 2013

References

Hebb DO. *The organization of Behavior. A Neuropsychological Theory*. Wiley. 1949

Aertsen AM, Gerstein GL, Habib MK, Palm G. *Dynamics of neuronal firing correlation: modulation of "effective connectivity"*. J Neurophysiol. 1989

Friston KJ, Frith CD, Liddle PF, Frackowiak RS. *Functional connectivity: the principal-component analysis of large (PET) data sets*. J Cereb Blood Flow Metab. 1993

Biswal B, Yetkin FZ, Haughton VM, Hyde JS. *Functional connectivity in the motor cortex of resting human brain using echo-planar MRI*. Magn Reson Med. 1995

Shulman GL, Fiez JA, Corbetta M, Buckner RL, Miezin FM, Raichle ME, Petersen SE. *Common Blood Flow Changes across Visual Tasks: 11. Decreases in Cerebral Cortex.* J Cognitive Neuroscience. 1997

Greicius MD, Krasnow B, Reiss AL, Menon V. *Functional connectivity in the resting brain: A network analysis of the default mode hypothesis*. PNAS. 2003

Medler DA. *A Brief history of connectionism*. Neural computing surveys. 1998

Callaway DS et al.. *Network Robustness and Fragility: Percolation on Random Graphs.* Physical Review Letters. 2000

Raichle ME, MacLeod AM, Snyder AZ, Powers WJ, Gusnard DA, Shulman GL. *A default mode of brain function*. Proc Natl Acad Sci U S A. 2001

Greicius MD, Krasnow B, Reiss AL, Menon V. *Functional connectivity in the resting brain: a network analysis of the default mode hypothesis*. Proc Natl Acad Sci U S A. 2003

References

Sporns O, Chialvo DR, Kaiser M, Hilgetag CC. *Organization, development and function of complex brain networks*. Trends Cogn Sci. 2004

Thadakamalla HP, Raghavan UN, Kumara S, Albert R. *Survivability of Multiagent-Based Supply Networks: A Topological Perspective*. IEEE Intelligent Systems. 2004

Sporns O, Chialvo DR, Kaiser M, Hilgetag CC. *Organization, development and function of complex brain networks*. Trends Cogn Sci. 2004

Beckmann CF, DeLuca M, Devlin JT, Smith SM. *Investigations into resting-state connectivity using independent component analysis.* Philos Trans R Soc Lond B Biol Sci. 2005

Borgatti SP, Everett MG. *A Graph-theoretic perspective on centrality.* Social networks. 2006

Raichle ME, Snyder AZ. *A default mode of brain function: A brief history of an evolving idea*. NeuroImage. 2007

Rogers BP, Morgan VL, Newton AT, Gore JC. *Assessing Functional Connectivity in the Human Brain by FMRI.* Magn Reson Imaging. 2007

Bullmore E, Sporns O. *Complex brain networks: graph theoretical analysis of structural and functional systems*. Nat Rev Neurosci. 2009

Van Den Heuvel MP, Hulshoff Pol HE. *Exploring the brain network: a review on resting-state fMRI functional connectivity*. Eur Neuropsychopharmacol. 2010

Lowe MJ. *A historical perspective on the evolution of resting-state functional connectivity with MRI*. Magn Reson Mater Phy. 2010

References

Baronchelli A, Ferrer-i-Cancho R, Pastor-Satorras R, Chater N, Christiansen MH. *Networks in Cognitive Science*. Trends in Cognitive Sciences. 2013

Hutchinson M et al.. *Dynamic functional connectivity: Promise, issues, and interpretations.* NeuroImage. 2013

Bordier C, Nicolini C, Bifone A. *Graph Analysis and Modularity of Brain Functional Connectivity Networks: Searching for the Optimal Threshold*. Frontiers in Neuroscience, 2017

Ting CM, Seghouane AK, Salleh SH, Noor AM. *Estimating Effective Connectivity from fMRI Data Using Factor-based Subspace Autoregressive Model.* IEEE Signal Processing Letters. 2014

Smitha KA , Raja KA, Arun KM , Rajesh PG, Thomas B, Kapilamoorthy TR, Kesavadas C. *Resting state fMRI: A review on methods in resting state connectivity analysis and resting state networks.* The Neuroradiology Journal. 2017

Vidaurre D, Smith SM, Woolrich MW. *Brain network dynamics are hierarchically organized in time*. PNAS. 2017

Michel CM, Koenig T. *EEG microstates as a tool for studying the temporal dynamics of whole-brain neuronal networks: A review.* NeuroImage. 2018

Demertzi et al.. *Human consciousness is supported by dynamic complex patterns of brain signal coordination.* Science advances. 2019

Shelke S, Attar V. *Source detection of rumor in social network – A review.* Online Social Networks and Media. 2019

Shahhosseini Y, Miranda MF. *Functional Connectivity Methods and Their Applications in fMRI Data*. Entropy. 2022